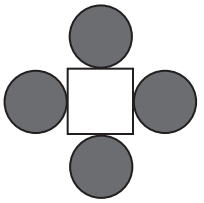
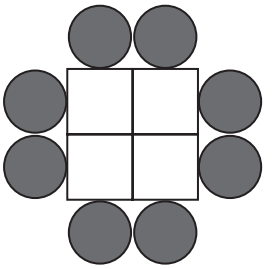
11 A sequence of patterns is made from circular tiles and square tiles

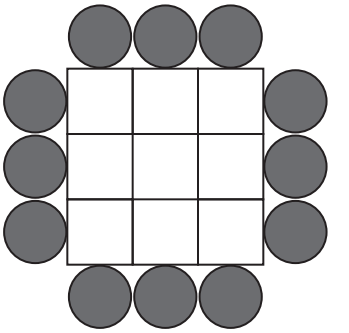
Here are the first three patterns in the sequence.



pattern number 1



pattern number 2



pattern number 3

(a) How many square tiles are needed to make pattern number 6?

Pattern n has n × n square tiles.

Pattern 6: 6 × 6 = 36

36

1 mark
Recognise square numbers

1 mark
Correct number

(2)

(b) How many circular tiles are needed to make pattern number 20?

There are n circles along each of 4 sides.

Pattern 20: 4 × 20 = 80

80

1 mark
4 circles per pattern number

1 mark
Correct number

(2)

Derek says,

"When the pattern number is odd, an odd number of square tiles is needed to make the pattern."

(c) Is Derek right?
You must give reasons for your answer.

Yes. The number of squares is n × n.

When n is odd, odd × odd is odd.

2 marks total
Conclusion supported by odd × odd

(2)

(Total for Question 11 is 6 marks)